

Appendices

Parallel Exact Top- k Vector Retrieval Engine

Course: Parallel and Distributed Algorithms (Pthreads, OpenMP, MPI) | **Student:** Abu Tayyab · 2023CS54

Instructor: Waqas Ali | **Submission Deadline:** 27 June 2026, 11:59 PM

APPENDIX A — PROJECT GANTT CHART

Project duration: 01 June 2026 – 27 June 2026 (27 days total)

#	Task	Start	End	Days	Est. Hours	Status
1	Environment Setup & Problem Analysis	01 Jun	04 Jun	3	6 h	✓ Done
2	Sequential Baseline — <code>main_seq.cpp</code>	04 Jun	06 Jun	2	10 h	✓ Done
3	Pthreads Implementation — <code>main_pth.cpp</code>	06 Jun	10 Jun	4	12 h	✓ Done
4	OpenMP Implementation — <code>main_omp.cpp</code>	10 Jun	13 Jun	3	8 h	✓ Done
5	MPI-Only Implementation — <code>main_mpi.cpp</code>	12 Jun	14 Jun	2	10 h	✓ Done
6	Hybrid MPI+OpenMP — <code>main_hybrid.cpp</code>	13 Jun	17 Jun	4	14 h	✓ Done
7	Benchmark Experiments & Data Collection	17 Jun	22 Jun	5	12 h	✓ Done
8	Paper Writing — Sections I–VI & References	10 Jun	26 Jun	16	20 h	✓ Done
9	Final Polish, Script Fix & Submission Packaging	26 Jun	27 Jun	1	4 h	✓ Done

Total estimated effort: 96 hours

Timeline Visualization

Each grid column represents approximately 2 days. Bars show active work periods.

Task	Jun 01	Jun 03	Jun 05	Jun 07	Jun 09	Jun 11	Jun 13	Jun 15	Jun 17	Jun 19	Jun 21	Jun 23
1. Setup & Analysis	Jun 01-04											

[illegible]

APPENDIX B — EXPERIMENTAL LOGBOOK

Hardware: Intel Core i5-1135G7, 4 cores / 8 logical threads, 2.40 GHz base / 4.20 GHz boost, 8 GB DDR4-3200, 6 MB L3 cache, 128 KB L1-D per core. OS: Ubuntu 22.04. Compiler: g++ -O3 (sequential, Pthreads, OpenMP) and mpicxx -O3 [-fopenmp] (MPI, Hybrid). Sequential reference: $T_{seq} = 0.5582$ s ($N = 200,000$, $D = 2,048$, $k = 5$).

B.1 — Dimensionality Scaling ($N = 100,000$, $k = 5$, 4 threads/ranks fixed)

Date & Time	Algorithm	Config	N (docs)	D (dims)	Runtime (s)	Speedup
21 Jun 18:30	Sequential	1×1	100,000	128	0.01747	1.00×
21 Jun 18:31	Pthreads	4 threads	100,000	128	0.00619	2.82×
21 Jun 18:31	OpenMP	4 threads	100,000	128	0.00537	3.25×
21 Jun 18:33	Sequential	1×1	100,000	1,536	0.20670	1.00×
21 Jun 18:34	Pthreads	4 threads	100,000	1,536	0.05581	3.70×
21 Jun 18:34	OpenMP	4 threads	100,000	1,536	0.05621	3.68×
21 Jun 18:35	Hybrid	$2r \times 2t$	100,000	1,536	0.05698	3.63×
21 Jun 18:37	Sequential	1×1	100,000	4,096	0.55663	1.00×
21 Jun 18:38	Pthreads	4 threads	100,000	4,096	0.14614	3.81×
21 Jun 18:38	OpenMP	4 threads	100,000	4,096	0.14817	3.76×

B.2 — Strong Scaling ($N = 200,000$, $D = 2,048$, $k = 5$)

Date & Time	Algorithm	Config	N (docs)	D (dims)	Runtime (s)	Speedup
26 Jun 20:12	Sequential	1×1	200,000	2,048	0.55821	1.00×
26 Jun 20:13	Pthreads	1 thread	200,000	2,048	0.58317	0.96×
26 Jun 20:13	Pthreads	2 threads	200,000	2,048	0.31176	1.79×
26 Jun 20:14	Pthreads	4 threads	200,000	2,048	0.27082	2.06×
26 Jun 20:14	Pthreads	8 threads	200,000	2,048	0.09782	5.71×
26 Jun 20:15	OpenMP	1 thread	200,000	2,048	0.56408	0.99×
26 Jun 20:15	OpenMP	2 threads	200,000	2,048	0.28261	1.98×
26 Jun 20:16	OpenMP	4 threads	200,000	2,048	0.14699	3.80×
26 Jun 20:16	OpenMP	8 threads	200,000	2,048	0.10533	5.30×
26 Jun 20:17	MPI-only	1 rank	200,000	2,048	0.57232	0.98×
26 Jun 20:17	MPI-only	2 ranks	200,000	2,048	0.37356	1.49×
26 Jun 20:18	MPI-only	4 ranks	200,000	2,048	0.25709	2.17×
26 Jun 20:18	MPI-only	8 ranks*	200,000	2,048	0.21106	2.64×

Date & Time	Algorithm	Config	N (docs)	D (dims)	Runtime (s)	Speedup
26 Jun 20:18	Hybrid	2r × 1t	200,000	2,048	0.37025	1.51×
26 Jun 20:19	Hybrid	2r × 2t	200,000	2,048	0.18190	3.07×
26 Jun 20:19	Hybrid	4r × 1t	200,000	2,048	0.27516	2.03×

* 8-rank MPI run used *mpirun --oversubscribe* on a 4-core machine.

B.3 — Weak Scaling (D = 2,048, k = 5, 50,000 documents per thread/rank)

Date & Time	Algorithm	Config	N (docs)	D (dims)	Runtime (s)	Δ vs. baseline
26 Jun 20:20	Pthreads	1 thread	50,000	2,048	0.15753	— (baseline)
26 Jun 20:20	Pthreads	2 threads	100,000	2,048	0.15182	−3.6 %
26 Jun 20:21	Pthreads	4 threads	200,000	2,048	0.14675	−6.8 %
26 Jun 20:21	Pthreads	8 threads	400,000	2,048	0.35664	+126.3 %
26 Jun 20:21	OpenMP	1 thread	50,000	2,048	0.19530	— (baseline)
26 Jun 20:22	OpenMP	2 threads	100,000	2,048	0.14994	−23.2 %
26 Jun 20:22	OpenMP	4 threads	200,000	2,048	0.20650	+5.7 %
26 Jun 20:22	OpenMP	8 threads	400,000	2,048	0.21680	+11.0 %
26 Jun 20:22	MPI-only	1 rank	50,000	2,048	0.13881	— (baseline)
26 Jun 20:23	MPI-only	2 ranks	100,000	2,048	0.17304	+24.7 %
26 Jun 20:23	MPI-only	4 ranks	200,000	2,048	0.22151	+59.6 %
26 Jun 20:23	MPI-only	8 ranks*	400,000	2,048	0.75928	+447.0 %

* 8-rank MPI weak-scaling run used *mpirun --oversubscribe*.

Total experiment runs logged: 38 distinct runs across 3 benchmark matrices.

★ **Best result: Pthreads 8 threads — 5.71× speedup (T = 0.09782 s vs. T_{seq} = 0.55821 s, N = 200,000, D = 2,048)**